

Why this, and what I hope to contribute

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Why this

I came to this project sideways. I was running nonlinear time-history analyses in OpenSeesPy for a coursework module on seismic design, and I found myself distracted by a simple question: how long each run took. A single five-story shear frame with moderate ground motion was taking around 50 milliseconds. That feels negligible until you think about what regional risk assessment actually requires — tens of thousands of buildings across a city, dozens of earthquake scenarios, multiple intensity levels. The arithmetic makes the problem clear quickly.

I knew neural networks were being applied to surrogate modelling in fluid mechanics and climate science under the label "Scientific ML," and I wondered whether the same idea could work for structural dynamics. What I didn't anticipate was how much the project would teach me about *why* specific modelling choices fail before they teach me how to fix them.

The three-story failure in EXP4 — where a Fourier Neural Operator achieved 99.6% error on buildings it had technically been trained on — was the most instructive moment. I had done something that looked reasonable: pad variable-height buildings to a common grid size. It took a systematic error analysis to trace back to why that was wrong (the Fourier transform assumes periodic boundaries; a hard step to zero is the worst possible violation of that assumption). Discovering and diagnosing that failure felt more valuable than if the model had just worked from the start.

I want to keep doing this kind of work — chasing the physical meaning of model failures, and trying to build representations that respect what the physics actually requires.

What I aim to do

SeismoFNO is a project that took longer than I expected, mostly because I kept finding things I couldn't explain yet. The modal phase-drift problem — where the model gets the peak displacement right but gets the frequency wrong — I still haven't solved. I know the mechanistic reason (global Fourier layers can't adapt their frequency basis dynamically), and I know the likely fix (state-space models or neural ODEs that process time causally), but building and evaluating that properly would take more time and expertise than I have in isolation right now.

That's precisely the situation where working with researchers who actually specialise in this matters. I want to spend a focused research period — ideally a full research project or internship — working on problems like this under supervision. Not to pad a CV, but because the intersection of structural dynamics and scientific machine learning is genuinely where I want to work.

What I wish to learn

The parts of this project I did worst were ablation design and uncertainty quantification. The shuffled-conditioning ablation in EXP6 was the closest I came to a proper falsification check, but it's blunt — shuffling within a batch doesn't test all the ways a model could be exploiting spurious correlations. I want to learn how to design ablations that are actually tight, and how to quantify confidence bounds on empirical results rather than just reporting point estimates.

More broadly, I want to work on problems where a result being wrong matters — where the output is used to make decisions about buildings or infrastructure, not just to beat a benchmark.

What I bring

A working end-to-end pipeline I built myself: OpenSeesPy ground-truth generation, graph neural operator architecture, physics-informed conditioning, a zero-leakage data split protocol, 305 passing unit tests, and frozen checkpoints with hash-verified evaluation CSVs. The ability to read structural dynamics literature and translate it into working code. A baseline of intellectual honesty about what doesn't work — the phase drift issue is documented as prominently in the repository as the results that look good.

I'm early in my training, so I'm not pretending to bring theoretical depth I don't have. What I can bring is a project that's real, a genuine interest in why it breaks where it does, and enough independence to be useful quickly rather than needing to be walked through the basics.

Repository: SeismoFNO (github.com/raghvendra1gdsc-png/SeismoFNO) · Full forensic audit, evaluation CSVs, and interactive demo available for inspection.